

Numerical Analysis PhD Exam, May 2005

Do any eight of the ten problems below.

1. This problem has three parts.

- Consider the matrix $A = uv^*$, where u and $v \in \mathbb{C}^n$. Under what condition on u and v is A a projector?
- Show that the Householder matrix $H = I - 2ww^*$, where $\|w\| = 1$, is unitary.
- Given a vector $x \in \mathbb{C}^n$ and an integer k with $1 < k < n$, derive a formula for a Householder matrix with the property that $(Hx)_i = 0$ for $i > k$ and $(Hx)_i = x_i$ for $i < k$. Be sure to choose the signs so that the formula is numerically stable.

2. This problem has three parts.

- Given a matrix $A \in \mathbb{C}^{m \times n}$ with n linearly independent columns, derive the formula (the normal equation) for the least squares solution to the possibly overdetermined linear system $Ax = b$.
- Given a matrix $A \in \mathbb{C}^{m \times n}$ with m linearly independent rows, derive the formula (the analogue of the normal equation) for the minimum norm solution $Ax = b$.
- Suppose you are given the singular value decomposition (SVD) $A = U\Sigma V^*$. Derive a single formula for the least squares/minimum norm solution to $Ax = b$ which is expressed in terms of U , Σ , and V .

3. Consider the invertible linear system $Ax = b$. Derive a tight bound for the condition number with respect to perturbations in b . If $A(x + \delta x) = b + \delta b$ is the perturbed system, then with the 2-norm, for which choice of b and for which perturbation δb is this bound tight?

4. Let

$$C = \begin{bmatrix} -a_1 & -a_2 & \cdots & -a_{n-1} & -a_n \\ 1 & 0 & \cdots & 0 & 0 \\ 0 & 1 & \ddots & \vdots & \vdots \\ \vdots & \ddots & \ddots & 0 & \vdots \\ 0 & \cdots & 0 & 1 & 0 \end{bmatrix}, \quad e = \begin{bmatrix} 1 \\ 0 \\ \vdots \\ \vdots \\ 0 \end{bmatrix}.$$

- What is the characteristic polynomial of C ?
- Consider any $A \in \mathbb{C}^{n \times n}$ and $b \in \mathbb{C}^n$. Let the characteristic polynomial of A be

$$p(\lambda) = \lambda^n + a_1\lambda^{n-1} + a_2\lambda^{n-2} + \cdots + a_{n-1}\lambda + a_n,$$

$$K = [b, Ab, A^2b, \dots, A^{n-1}b], \quad R = \begin{bmatrix} 1 & a_1 & a_2 & \cdots & a_{n-1} \\ 0 & 1 & a_1 & \cdots & a_{n-2} \\ \vdots & \ddots & \ddots & \ddots & \vdots \\ \vdots & & \ddots & \ddots & a_1 \\ 0 & \cdots & \cdots & 0 & 1 \end{bmatrix}.$$

Show that if K is invertible, then

$$(KR)^{-1}A(KR) = C \quad \text{and} \quad (KR)^{-1}b = e.$$

(Suggestion: Prove that $A(KR) = (KR)C$ using $p(A) = 0$.)

5. Let $A = QR$ be the factorization of A into the product of a unitary matrix and a triangular matrix. Suppose that the columns of A are linearly independent. Show that $|r_{kk}|$ is the distance from the k -th column of A to the linear space spanned by the first $k - 1$ columns of A .
6. Show that $x = \pi + 0.5 \sin(x/2)$ has a unique solution α in $\mathbb{R}$. Give a fixed point iteration by which one can compute an approximation to α numerically and prove that it converges at order one.
7. The Lagrange interpolation problem is to find a polynomial $p_n(x)$ of degree at most n such that

$$p_n(x_i) = y_i, \quad i = 0, \dots, n,$$

where $x_0, \dots, x_n$ are distinct real points and $y_0, \dots, y_n$ are given data. Prove that this problem always has a unique solution.

8. Consider the quadrature that approximates

$$\int_a^b f(x) dx$$

by

$$I_3(f) := \int_a^b P_3(x) dx,$$

where $P_3(x)$ is the cubic Hermite interpolant to $f(x)$ at the points $x = a, b$. Obtain an error formula for this approximation. (This is called the corrected trapezoid rule.)

9. Consider multistep methods of the form

$$y_{n+1} = y_{n-q} + h \sum_{j=-1}^p b_j f(x_{n-j}, y_{n-j})$$

with $q \geq 1$. Show that such methods do not satisfy the strong root condition. Find an example with $q = 1$ that is relatively stable.

10. Let S_n^1 denote the space of linear splines based on knots $a = t_0 < t_1 < \dots < t_n = b$, i.e.,

$$S_n^1 = \left\{ v : v|_{[t_j, t_{j+1}]} \text{ is linear for all } j = 0, 1, \dots, n-1 \text{ and } v \text{ is continuous on } [a, b] \right\}.$$

Let $s_f \in S_n^1$ interpolate a continuous function f at the knots.

- (a) Prove that $\|s_f\|_\infty \leq \|f\|_\infty$. (Here $\|v\|_\infty = \max_{x \in [a, b]} |v(x)|$.)
- (b) Prove that $\|f - s_f\|_\infty \leq 2\|f - s\|_\infty$ for any $s \in S_n^1$.
- (c) Consider the error in best approximation to f from S_n^1 and the error in approximating f by its linear interpolant s_f . If one goes to zero, does the other go to zero?